

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY BHOPAL

Department of Computer Science and Engineering

Branch: - CSE

Semester - I

Mid-Term Examination October - 2025

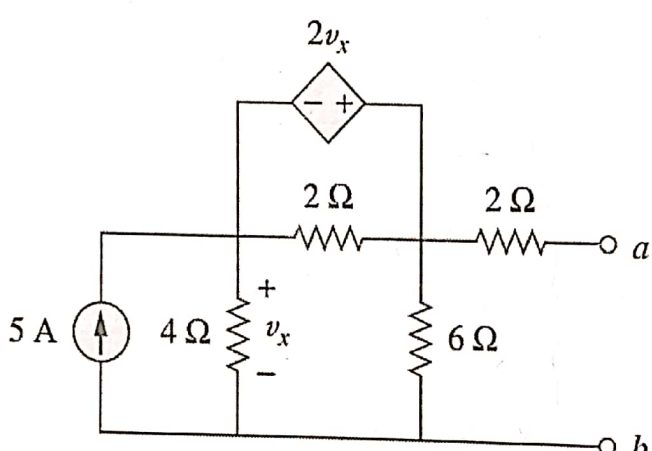
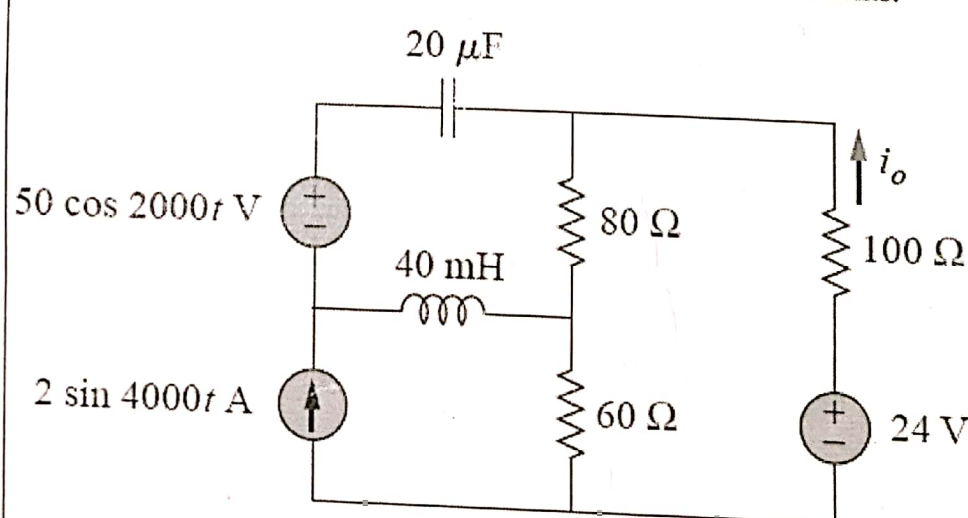
Subject Name/ Code: - Fundamentals of Electrical and Electronics (CSE-1004)

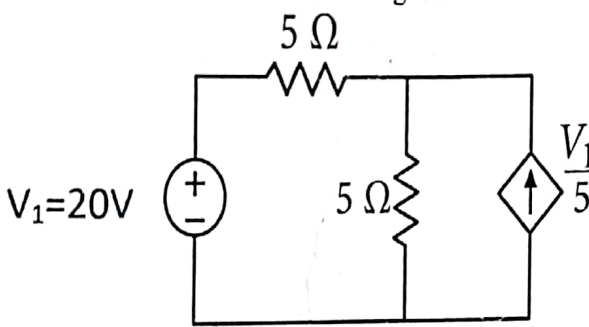
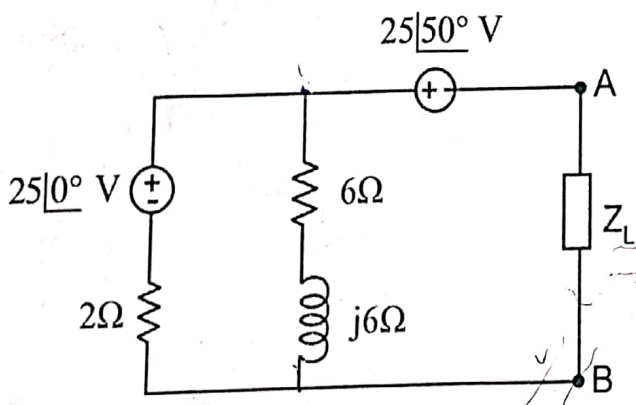
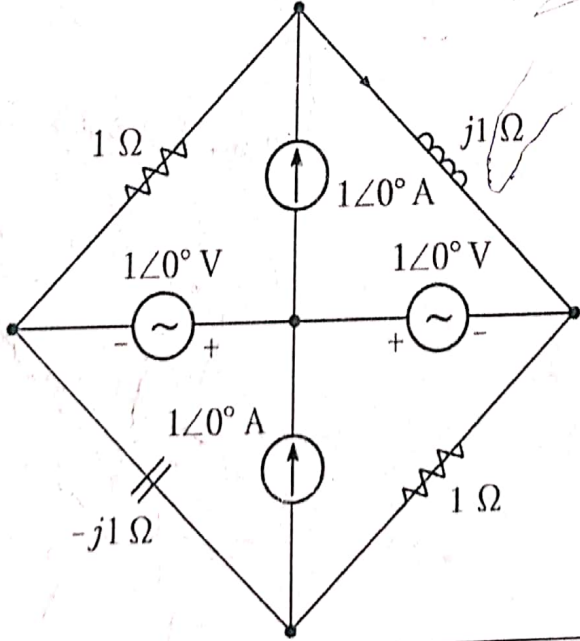
Max. Marks: 20

Time Allotted: 90 min

NOTE: 1. All questions are compulsory.

2. Mark answers number clearly.

S. No	Question	Marks
1.	Analyze the given circuit containing an independent current source, a dependent voltage source, and load resistors, and determine the Thevenin and Norton equivalent circuits across terminals a and b. 	4
2.	Apply the superposition theorem to calculate the current i_o in the given circuit consisting of voltage and current sources with RLC elements. 	4

3.	Analyze the given circuit and find the power deliver/absorb by the dependent current source shown in the figure. 	4
4.	Apply the maximum power transfer theorem to find the load impedance (Z_L) and the maximum power delivered to the load in the given circuit. 	4
5.	In the circuit shown below, calculate the current through the inductor. 	4